Android 360° Photo Sphere Capture Protocol

Fundamentals: App Options, Nodal Pivot Physics & 3-Step Viewfinder Alignment

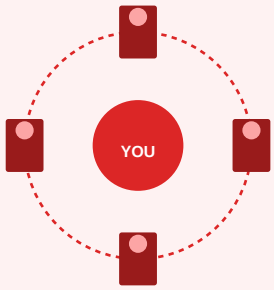
1. Recommended Android Applications (No Dedicated 360 Camera Required)

Application	Compatibility	Key Features & How to Launch
Google Camera (GCam)	Pixel native / GCam port on Samsung, OnePlus, Xiaomi	Select **Modes** → **Photo Sphere**. Automatically snaps frames as crosshair reaches floating target dots.
Panorama 360 & VR	Google Play Store (by TeliportMe)	Universal Android support. Real-time gyroscope stitching, intuitive leveling guides, and equirectangular export.
Photosphere 360	Google Play Store	Lightweight, zero-lag interface with automated shutter trigger on dot-alignment. Direct 2:1 JPEG output.

2. The Golden Nodal Point Rule (Eliminating Stitching Faults & Parallax)

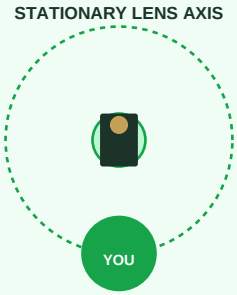
Parallax distortion happens when nearby foreground objects shift against the background between shots. To prevent split pillars and broken stage steps, the phone lens must remain anchored to an imaginary stationary pin:

[X] WRONG: Arms Swing in Wide Circle



Lens moves in 3-ft radius: Stage pillars, tables & lights get cut in half

[OK] CORRECT: Photographer Pivots Around Phone

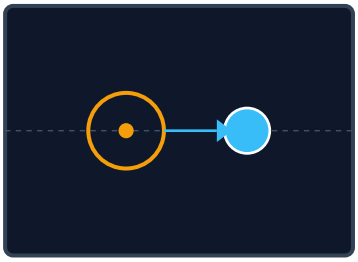


Lens stays locked in 1 spot: Zero parallax displacement, flawless stitch

3. Viewfinder Capture System: Step-by-Step Dot Snapping UI

The app guides you continuously through floating target points. Understand the 3-step capture cycle:

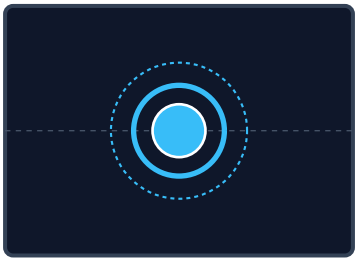
STEP 1: LOCATE BLUE DOT



Center reticle is empty.

Rotate your body slowly toward blue dot.

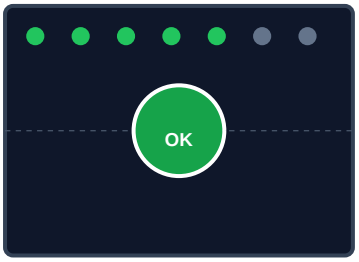
STEP 2: RETICLE ALIGNMENT



Target dot enters center ring.

Hold phone completely still for 0.5s.

STEP 3: SHUTTER & PROGRESS



Camera clicks automatically!

New dot appears. Follow arrows to next.

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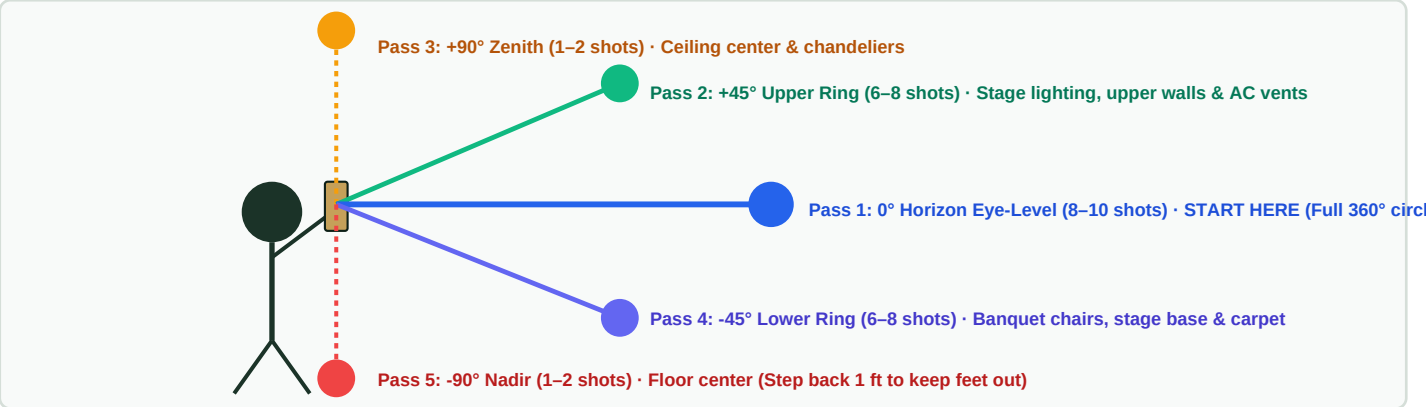
Spherical Elevation Protocol & Venue Capture Map at Thekkanath Souparnika

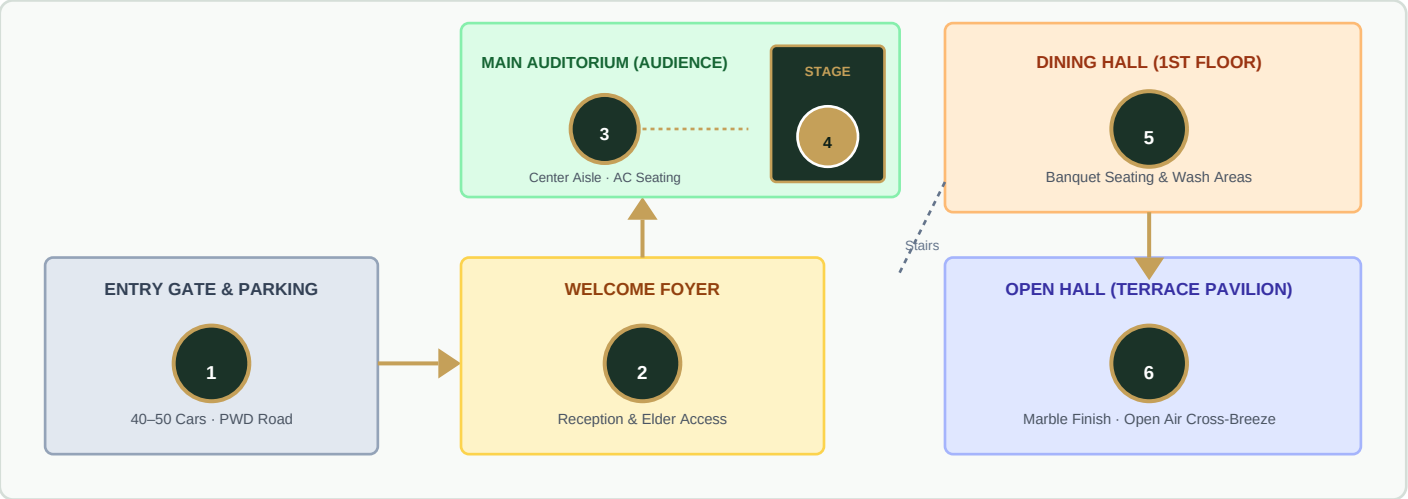
4. The 5-Ring Vertical Elevation Protocol (Complete 360° × 180° Sphere)

A full spherical panorama requires 5 distinct rings of photos from the exact same standing spot. Complete them in order:



5. Venue Architectural Layout & 6 Vantage Nodes (Walkthrough Map)

Follow this sequential navigation path through Thekkanath Souparnika to ensure seamless virtual tour connections:



#	Node Location	Exact Standing Spot	Key Features to Capture
1	**Compound Front Gate**	Center of front gate looking into parking lot	40–50 car parking capacity, building facade, approach road.
2	**Welcome Hall Foyer**	Just inside main glass entrance doors	Reception desk, side sofa seating, ground floor step-free access.
3	**Stage Hall (Audience)**	Center aisle, 15 ft back from stage	Full decorated stage, warm LED lights, high ceiling AC.
4	**On the Stage**	Center stage looking out over auditorium	Host/VIP view across audience chairs and double entrance doors.
5	**Dining Hall**	Center aisle between rows of banquet tables	Long dining tables, window garden views, handwash counters.
6	**Open Hall Pavilion**	Center floor of covered rooftop terrace	Marble-finish flooring, ceiling fans, open perimeter ventilation.

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Field Best Practices, 2:1 Format Verification & Handover Instructions

6. Practical Field Rules: Critical DOs & DON'Ts

Follow these 4 practical field rules to avoid blurry photos, exposure mismatches, and ghost artifacts:

OK

CLEAN CAMERA LENS

Wipe lens with clean cloth.
Skin oil causes heavy glare
around stage floodlights.

OK

TURN ALL LIGHTS ON

Switch on chandeliers, LEDs,
and decorative wall sconces.
Avoid half-dark halls.

NO

NO MOVING PEOPLE

Ask staff to remain outside.
Moving bodies create ghost
clones & cut-off limbs.

OK

5-FT WALL DISTANCE

Never stand right next to
pillars or door frames.
Keep 5–6 ft open buffer.

7. Output Format: Equirectangular 2:1 Projection & Web Tour Wrapping

When finished, the app renders a flat rectangular image called an **Equirectangular Projection** with an exact **2:1 aspect ratio** (width = 2 × height):

FLAT EQUIRECTANGULAR JPEG (2:1 RATIO, e.g. 8192 × 4096 px)

WebGL Wraps

INTERACTIVE 3D VIRTUAL TOUR

FILE SPECIFICATIONS:

- Format: High-res JPEG (.jpg)
- Aspect Ratio: Exactly 2:1
- Resolution: 4096×2048 or 8192×4096
- File Size: ~8 MB – 18 MB per node

8. Quality Transfer & Handover to Web Team

[CRITICAL RULE] DO NOT SEND VIA REGULAR WHATSAPP PHOTO SHARING
Standard WhatsApp photo sharing applies heavy lossy compression, shrinking images to 1600px and stripping 360° equirectangular metadata tags. The virtual tour will look degraded and pixelated.
[APPROVED HIGH-RESOLUTION TRANSMISSION CHANNELS]
1. WhatsApp 'Document' Mode: In WhatsApp chat, tap the Attachment icon (paperclip) → Choose **Document** → Browse and select the original .jpg files from internal storage.
2. Google Drive / WeTransfer: Upload full uncompressed files to a Drive folder and share link with thekkanathsouparnika@gmail.com.
3. Direct USB / PC Transfer: Connect Android device to computer via USB cable, open DCIM/Camera/ or Pictures/Panoramas/, and copy raw original JPEGs.
When received, photos are placed into public/assets/tour/ and the interactive WebGL 360° tour route is deployed instantly!